

Information Sheet FMAN 323-3

Effects of the Economic Environment on Business Strategy

Learning Objectives:

After reading this INFORMATION SHEET, YOU MUST be able to:

- 1. Explore the importance of different methods and analysis used in business to make sound decisions.
- 2. Understand the role of financial markets and institutions

Effects of the Economic Environment on Business Strategy



Introduction

Financial markets play a vital role in facilitating the smooth operation of capitalist economies by allocating resources and creating liquidity for businesses and entrepreneurs. The markets make it easy for buyers and sellers to trade their financial holdings. Financial markets create securities products that provide a return for those who have excess funds (Investors/lenders) and make these funds available to those who need additional money (borrowers).

	Bachelor of Science in Accounting Information System FINANCIAL MANAGEMENT- FMan223	Bulacan Polytechnic College	Date Developed: January 2021 Date Revised:	Page 3 of 19
		Document No. c/o Admin	Developed by: Mr. Eugene A. Ruano	Revision # 00

ROLE OF FINANCIAL MARKETS AND INSTITUTIONS

Financial markets, such as bond and stock markets, are crucial in our economy.

Well-functioning financial markets, such as the *bond market*, *stock market*, and *foreign exchange market*, are key factors in producing high economic growth.

Why Study Financial Markets?

- These markets channel funds from savers to investors, thereby promoting economic efficiency.
- Market activity affects personal wealth, the behavior of business firms, and economy as a whole

Debt markets, or bond markets, allow governments, corporations, and individuals to borrow to finance activities.

In this market, borrowers issue a security, called a bond, that promises the timely payment of interest and principal over some specific time horizon.

The interest rate is the cost of borrowing



The **stock market** is the market where common stock (or just stock), representing ownership in a company, are traded.

Companies initially sell stock (in the primary market) to raise money. But after that, the stock is traded among investors (secondary market).

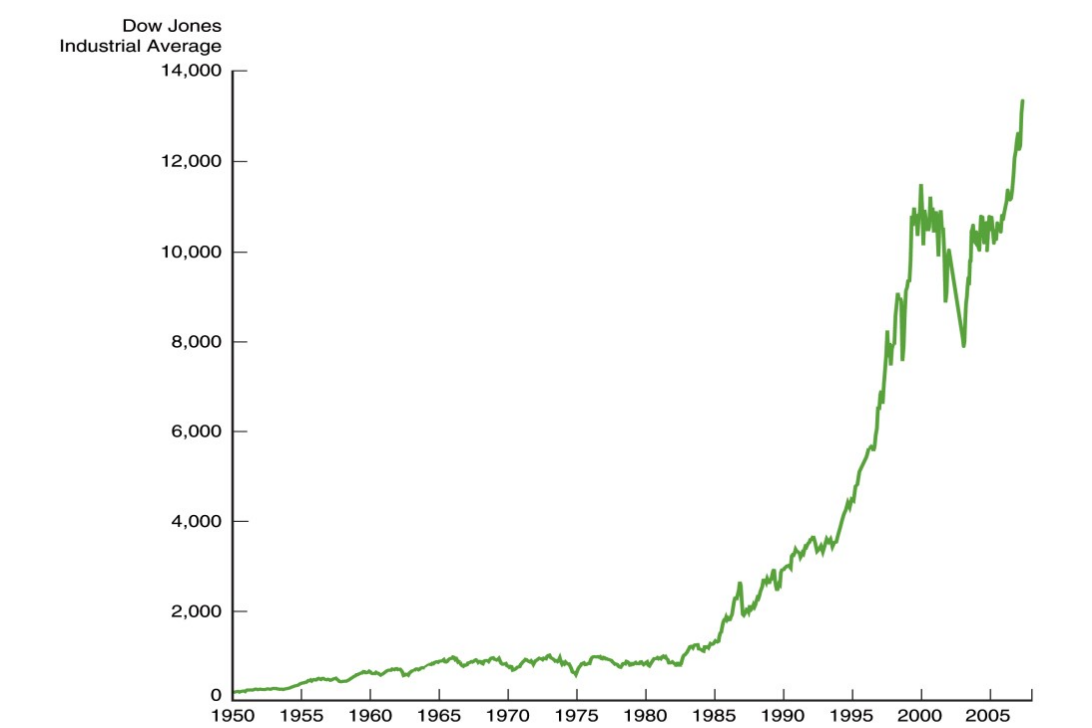


Figure 1.2 Stock Prices as Measured by the Dow Jones Industrial Average, 1950–2007

Source: Dow Jones Indexes: <http://finance.yahoo.com/?u>.

The Foreign Exchange Market

The foreign exchange market is where international currencies trade and exchange rates are set.

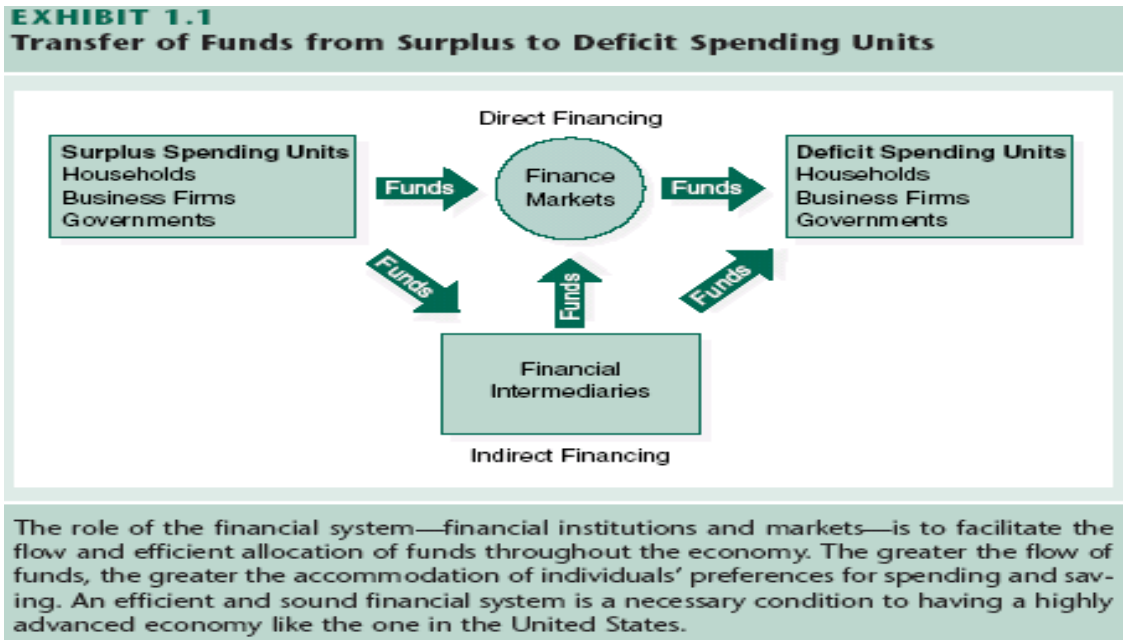


FUNCTIONS OF FINANCIAL MARKET

A **Financial Market** is a market in which financial assets (securities) can be purchased or sold.

Financial markets facilitate transfers of funds from person or business without investment opportunities (i.e., “Lender-Savers”, or “Surplus Unit”) to those who have investment opportunities (i.e., “Borrower-Spenders”, or “Deficit Unit”).

Segments of Financial Markets



Direct Financing

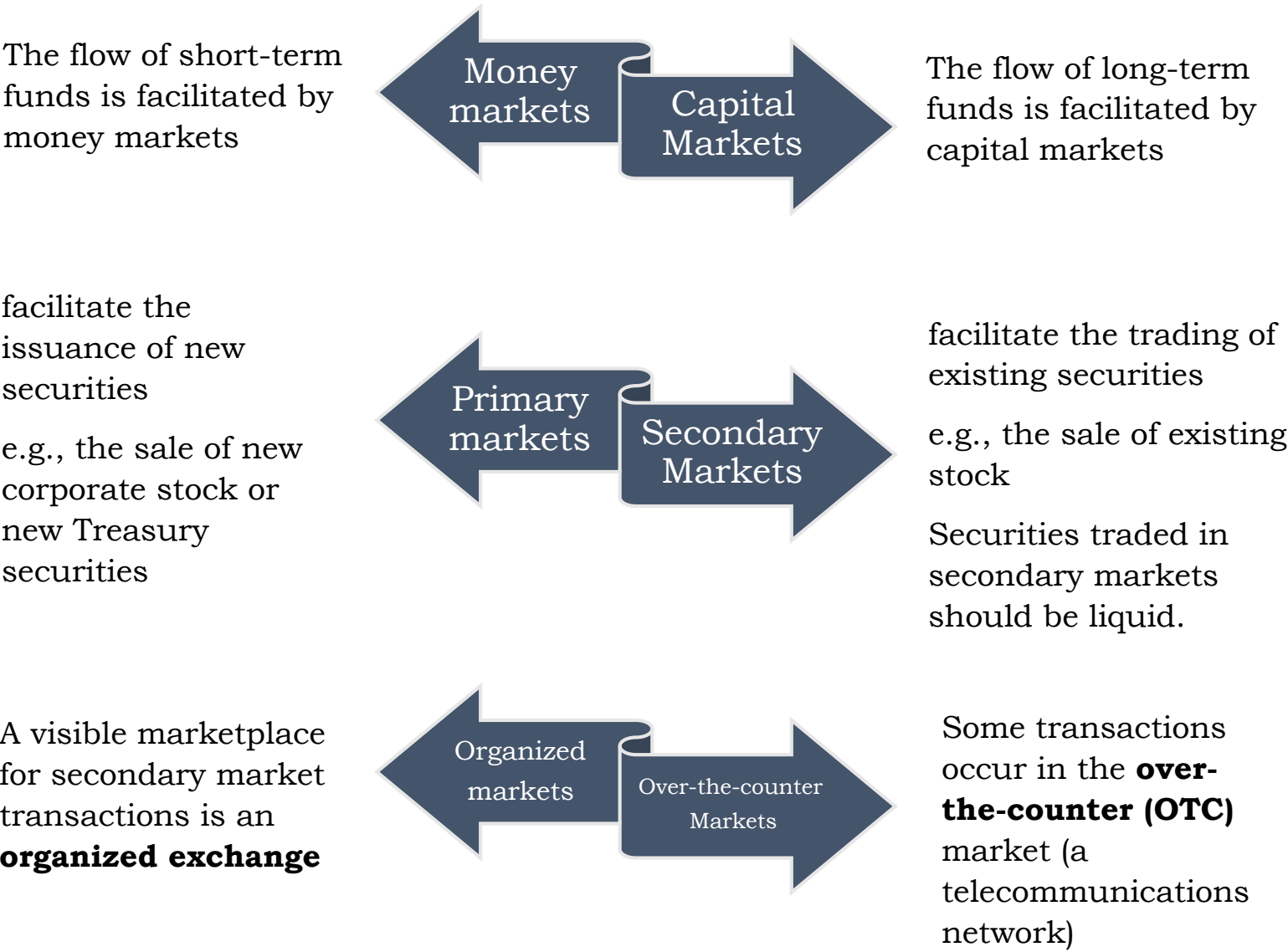
- Funds are transferred directly from ultimate savers to ultimate borrowers

Indirect Financing

- A financial "intermediary" transforms financial claims with one set of characteristics into financial claims with other characteristics e.g. deposits are used to make loans.

TYPES OF FINANCIAL MARKETS

Financial markets can be distinguished by the maturity structure and trading structure of its securities



Knowledge of financial markets is power:

- ✓ Decide which markets to use to achieve our investment goals or financing needs
- ✓ Decide which markets to use as part of your job
- ✓ Avoid common mistakes in investing and borrowing

ROLE OF FINANCIAL INSTITUTIONS IN FINANCIAL MARKETS

Role of depository institutions 	Depository institutions accept deposits from surplus units and provide credit to deficit units. Depository institutions are popular because: - Deposits are liquid - They customize loans - They accept the risk of loans - They have expertise in evaluating creditworthiness - They diversify their loans
Commercial Banks 	are the most dominant depository institution. Offer a wide variety of deposit accounts. Transfer deposited funds by providing direct loans or purchasing debt securities. Serve both the public and the private sector.

Savings Institutions 	Include savings and loan associations (S&Ls) and savings banks. Are mostly owned by depositors (mutual). Concentrate on residential mortgage loans.
Credit Unions 	Credit unions are nonprofit organizations. Restrict their business to credit union members. Tend to be much smaller than other depository institutions.
Role of Non depository Financial Institutions	Non depository institutions generate funds from sources other than deposits Finance companies - Obtain funds by issuing securities - Lend funds to individuals and small businesses
Mutual Funds 	Sell shares to surplus units. Use funds to purchase a portfolio of securities - Some focus on capital market securities (e.g., stocks or bonds) - Money market mutual funds concentrate on money market securities

Securities firms 	Broker function  In
Insurance Companies 	Provide insurance policies to individuals and firms for death, illness, and damage to property. Charge premiums. Invest in stocks or bonds issued by corporations.
Pension Funds 	Offered by most corporations and government agencies. Manage funds until they are withdrawn from the retirement account. Invest in stocks or bonds issued by corporations or in bonds issued by the government.

International trade between different countries is an important factor in raising living standards, providing employment and enabling consumers to enjoy a greater variety of goods.

International trade has occurred since the earliest civilizations began trading, but in recent years international trade has become increasingly important with a larger share of GDP devoted to exports and imports



World Bank stats show how world exports as a % of GDP have increased from 12% in 1960 to around 30% in 2015.

With an increased importance of trade, there have also been growing concerns about the potential negative effects of trade – in particular, the unbalanced benefits with some losing out, despite overall net gains.

Importance of trade

- ✚ Make use of abundant raw materials
- ✚ Comparative advantage
- ✚ Greater choice for consumers
- ✚ Specialization and economies of scale – greater efficiency
- ✚ Service sector trade.
- ✚ Global growth and economic development

TIME VALUE OF MONEY

 Bachelor of Science in Accounting Information System FINANCIAL MANAGEMENT- FMan223	Bulacan Polytechnic College	Date Developed: January 2021 Date Revised:	Page 11 of 19
	Document No. c/o Admin	Developed by: Mr. Eugene A. Ruano	Revision # 00

$$\text{Future Money} = \text{Present Money} + \text{Interest}$$


The time value of money is used to determine whether future benefits are sufficiently large to justify current outlays

Concept of Interest

Simple Interest

- is determined by multiplying the daily interest rate by the principal by the number of days that elapse between payments. It is given only once.

$$I = Prt$$

where:

I = Interest

P = Principal amount

r = Annual interest rate

t = time, in years

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

where:

A = future value

P = Principal amount (initial value)

r = interest rate

t = time

n = number of times compounded in one "t"

Compound Interest

- is the interest on a loan or deposit calculated based on both the initial principal and the accumulated interest from previous periods.

Mathematical tools of the time value of money are used in making capital allocation decisions

Assuming that the worth of \$1,000 needs to be calculated after 4 years at a 10% interest per year, we have:

$$1^{\text{st}} \text{ year} \quad \$1,000 \times 1.10 = \$1,100$$

$$2^{\text{nd}} \text{ year} \quad \$1,100 \times 1.10 = \$1,210$$

$$3^{\text{rd}} \text{ year} \quad \$1,210 \times 1.10 = \$1,331$$

$$4^{\text{th}} \text{ year} \quad \$1,331 \times 1.10 = \$1,464$$

 Bachelor of Science in Accounting Information System FINANCIAL MANAGEMENT- FMan223	Bulacan Polytechnic College		Date Developed: January 2021 Date Revised:	Page 12 of 19
	Document No. c/o Admin	Developed by: Mr. Eugene A. Ruano	Revision # 00	

Future Value – Single Amount

A generalized formula for Future Value:

Where:

$$FV = PV(1 + i)^n$$

FV = Future value

PV = Present value

i = Interest rate

n = Number of periods;

In the previous case, PV = \$1,000, i = 10%, n = 4, hence;

FV = \$1,000(1.10)⁴, or \$1,000 × 1.464 = \$1,464

Periods	1%	2%	3%	4%	6%	8%	10%
1	1.010	1.020	1.030	1.040	1.060	1.080	1.100
2	1.020	1.040	1.061	1.082	1.124	1.166	1.210
3	1.030	1.061	1.093	1.125	1.191	1.260	1.331
4	1.041	1.082	1.126	1.170	1.262	1.360	1.464
5	1.051	1.104	1.159	1.217	1.338	1.469	1.611
10	1.105	1.219	1.344	1.480	1.791	2.159	2.594
20	1.220	1.486	1.806	2.191	3.207	4.661	6.727

In determining future value, the following can be used:

$$FV = PV \times FV_{IF}$$

Where: FV_{IF} = the interest factor

Present Value – Single Amount

A sum payable in the future is worth less today than the stated amount

The formula for the present value is derived from the original formula for future value:

$$FV = PV(1 + i)^n$$
Future value

$$PV = FV \left[\frac{1}{(1 + i)^n} \right]$$
Present value

Future Value – Annuity

Annuity:

- A series of consecutive payments or receipts of equal amount

Future Value of an Annuity:

- Calculated by compounding each individual payment into the future and then adding up all of these payments

A generalized formula for Future Value of Annuity:

$$FV_A = A \times FV_{IFA}$$

Where:

FV_A = Future value of the Annuity

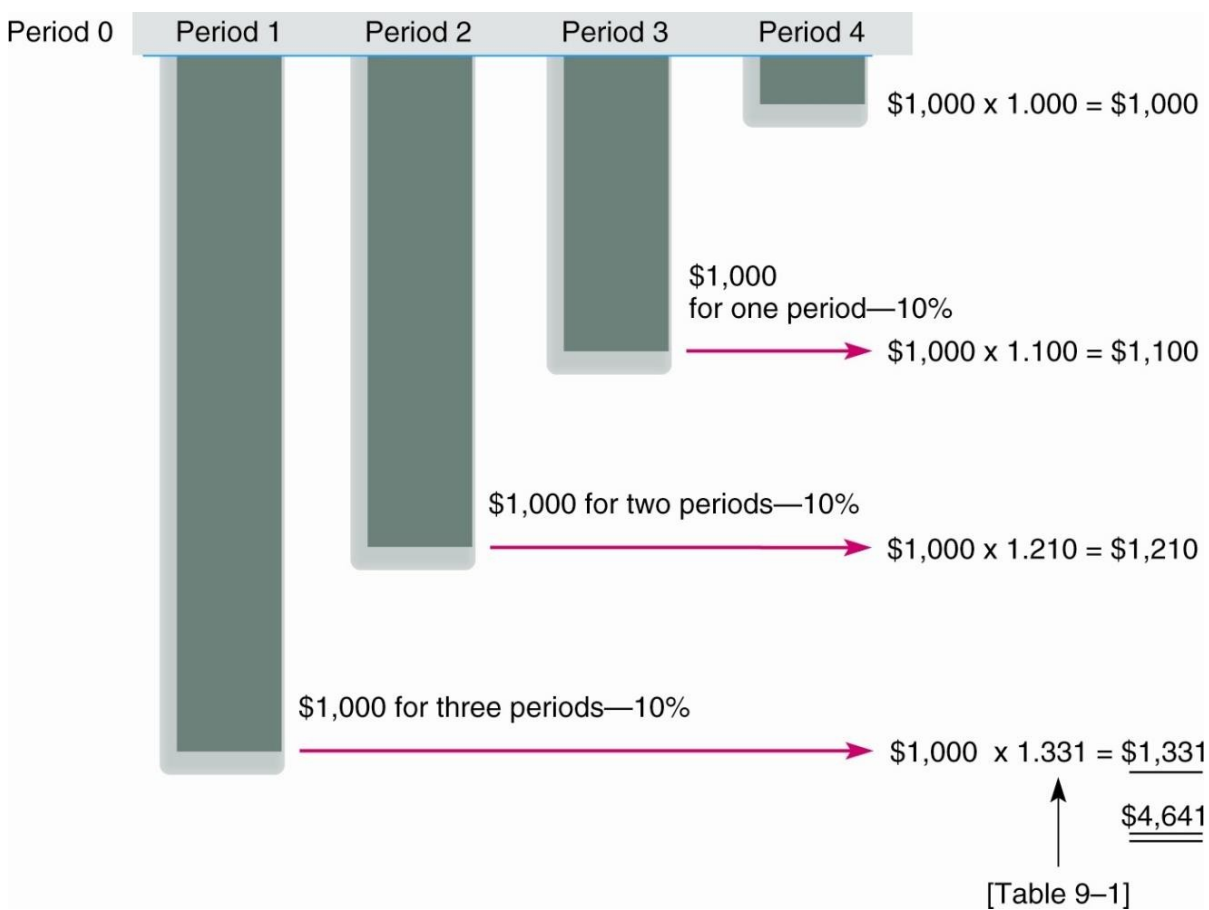
FV_{IFA} = Annuity Factor = $\{[(1+i)^n - 1] \div i\}$

A = Annuity value

i = Interest rate

n = Number of periods;

Compounding Process for Annuity



Future Value of an Annuity of \$1(FV_{IFA})

Periods	1%	2%	3%	4%	6%	8%	10%
1	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2	2.010	2.020	2.030	2.040	2.060	2.080	2.100
3	3.030	3.060	3.091	3.122	3.184	3.246	3.310
4	4.060	4.122	4.184	4.246	4.375	4.506	4.641
5	5.101	5.204	5.309	5.416	5.637	5.867	6.105
10	10.462	10.950	11.464	12.006	13.181	14.487	15.937
20	22.019	24.297	26.870	29.778	36.786	45.762	57.275
30	34.785	40.588	47.575	56.085	79.058	113.280	164.490

An expanded table is presented in Appendix C.

Present Value – Annuity

Calculated by discounting each individual payment back to the present and then adding up all of these payments

A generalized formula for Present Value of Annuity:

$$PV_A = A \times PV_{IFA}$$

Where:

PV_A = Present value of the Annuity

PV_{IFA} = Annuity Factor = {1 – [1 ÷ (1+i)ⁿ] ÷ i}

A = Annuity value

i = Interest rate

n = Number of periods

Assuming that A = \$1,000, n = 4, i = 10%, we have:

$$PV_A = A \times PV_{IFA} \quad (n = 4, i = 10\%)$$

$$PV_A = \$1,000 \times 3.170 = \$3,170$$

Periods	1%	2%	3%	4%	6%	8%	10%
1	0.990	0.980	0.971	0.962	0.943	0.926	0.909
2	1.970	1.942	1.913	1.886	1.833	1.783	1.736
3	2.941	2.884	2.829	2.775	2.673	2.577	2.487
4	3.902	3.808	3.717	3.630	3.465	3.312	3.170
5	4.853	4.713	4.580	4.452	4.212	3.993	3.791
8	7.652	7.325	7.020	6.773	6.210	5.747	5.335
10	9.471	8.983	8.530	8.111	7.360	6.710	6.145
20	18.046	16.351	14.877	13.590	11.470	9.818	8.514
30	25.808	22.396	19.600	17.292	13.765	11.258	9.427

An expanded table is presented in Appendix D.

Deferred Annuity

Situations involving a combination of single amounts and an annuity.

When annuity is paid sometime in the future

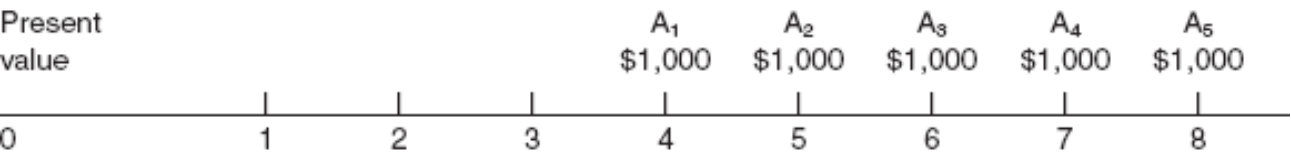
Assuming a contract involving payments of different amounts each year for a three-year period:

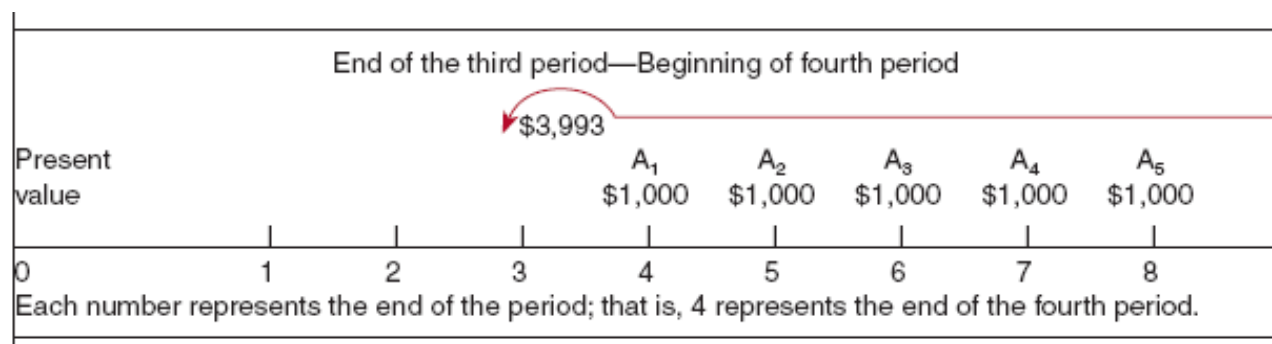
An annuity of \$1,000 is paid at the end of each year from the fourth through the eighth year

To determine the present value of the cash flows at 8% discount rate

1. \$1,000	} Present value = \$5,022
2. 2,000	
3. 3,000	
4. 1,000	} Five-year annuity
5. 1,000	
6. 1,000	
7. 1,000	
8. 1,000	

To determine the annuity





To discount the \$3,993 back to the present, which falls at the beginning of the fourth period, in effect, the equivalent of the end of the third period, it is discounted back three periods, at 8% interest rate

$$PV = FV \times PV_{IF} \qquad (n = 3, i = 8\%)$$

$$PV = \$3,993 \times 0.794 = \$3,170 \text{ (actual present value)}$$

